

Figure 6: Installation of FreeNAS OS is successfully completed. Say OK

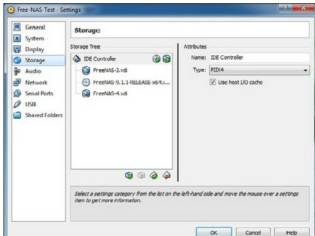


Figure 7: Snapshot of FreeNAS vm storage disk configuration

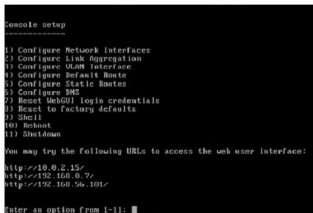


Figure 8: FreeNAS boot screen

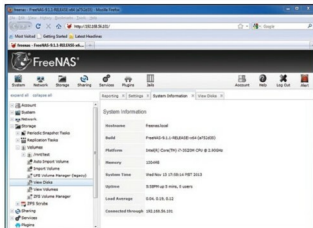


Figure 9: FreeNAS webGUI for admin tasks

could be of interest to us:

1. Along with sharing the disk through NFS, FreeNAS can also share the storage through iSCSI.
2. It supports many common file sharing protocols like AFP, CIFS, FTP, NFS, SFTP and TFTP.
3. It supports LDAP for user authentication.
4. It supports the creation of UFS2-based volumes including *gmirror*, *gstripe* and *graid3*.
5. It supports the creation of ZFS volume.
6. Just like any other proprietary storage solution,

FreeNAS also provides features like replication of data, data protection using RAID, regular back-up services, encryption of data, easy-to-use GUI Web tools—the list keeps growing. For many more features, please explore the FreeNAS website listed in the 'References' section.

Now, as we already know the features of FreeNAS, it's very easy to think up use cases for it, depending upon your need. It can play a role in households that have only a single system with all the movies and pictures saved in one place. A similar role for FreeNAS can be envisaged in hostels where sharing is very common among friends. Coming to more serious use cases, it can be used to share documents across a group of people.

As this solution is open source and covers the breadth of various subsystems of an operating system, it provides unlimited opportunity for developers to be a part of the community according to their areas of interest. END 

Some common FreeNAS mailing lists and communities

1. Community announcements: <http://forums.freenas.org/forums/announcements.27/>
2. FreeNAS free help and support for all your queries: <http://forums.freenas.org/forums/help-support.2/>
3. Developers discussion forum: <http://forums.freenas.org/forums/development.8/>
4. FreeNAS how-to documentations: <http://forums.freenas.org/forums/how-to-guides.13/>
5. A few FreeNAS mailing lists are listed here and you are free to be a part of them: http://doc.freenas.org/index.php/Mailing_Lists

References

The Linux Kernel Module Programming Guide

- [1] <http://www.freenas.org>
- [2] <http://compnetworking.about.com>
- [3] http://www.freenas.org/images/resources/freenas9.1.1/freenas9.1.1_guide.pdf

By: Ashish Bunkar

The author works as a Linux engineer at the Dell India R&D Centre, Bengaluru, and has more than three years of experience in Linux and UNIX-based operating systems and applications. He is interested in SAN and virtualisation technologies.